ChatGPT

What is the clinical significance of PIK3R2 G373R in CIViC?

Understood! I'll investigate the clinical significance of the PIK3R2 G373R variant in the CIViC database to find the latest information. I'll gather relevant data from updated sources and share the detailed findings with you soon.

Worked for 5 minutes >

The CIViC evidence item for **PIK3R2 G373R** (EID9420) classifies this variant as a **gain-of-function** change. In the supporting study, lymphoblastoid cell lines from patients with megalencephaly syndromes who carried the *PIK3R2* p.G373R mutation were compared with control cells using phosphatidylinositol-3,4,5-trisphosphate (PIP3) immunostaining. Cells with the G373R mutation had **higher PIP3 levels** than control cells, indicating increased PI3K activity; inhibition of PI3K with PI-103 reduced PIP3 levels in the patient cell line civicdb.org. CIViC therefore classifies this variant as a **gain-of-function** alteration and notes it as a **rare germline** mutation with evidence level D (pre-clinical) civicdb.org. MedlinePlus also describes the Gly373Arg (G373R) mutation as the most common PIK3R2 change and states that it increases PI3K activity, promoting signaling in the PI3K-AKT-mTOR pathway and leading to excessive cell growth medlineplus.gov.

Clinical significance in CIViC

- **Variant type:** rare germline missense change in the *PIK3R2* gene (p.Gly373Arg).
- **Functional classification:** gain-of-function – mutant cells showed increased PIP3 levels and enhanced PI3K activity; PI3K inhibition reversed this effect civicdb.org.
- **Evidence level & context:** level D (pre-clinical) evidence from Rivière et al., 2012 civicdb.org. The evidence is functional rather than disease-specific, so there are currently no therapy or disease associations listed for this variant in CIViC.

Thus, in CIViC the PIK3R2 G373R variant is recognized as a gain-of-function germline variant that enhances PI3K signaling; this assessment is based on pre-clinical functional studies rather than clinical outcome data.

Sources

Describe a task

+ Agent Sources